HACK FOR SOCIAL IMPACT 2026 · UNCCD CHALLENGE

Neutral Ground

A real storm is coming. You have three autumns to get a San Francisco canyon ready.

A 6–8 minute browser game set in the real Glen Canyon Park, played through its real 2020–2023 weather



WHY A GAME

Young people care about the land, and many feel powerless

59%

of 10,000 people aged 16–25 in ten countries were very or extremely worried about climate change

Over half

said they felt powerless and helpless about it

Our answer: agency

In Neutral Ground your own choices, in a real place, change what its real weather does. You see it happen, then try again.

Source: Hickman et al., “Climate anxiety in children and young people”, The Lancet Planetary Health, 2021

THE IDEA

A real canyon, its real weather, your choices

Real place

Glen Canyon Park, San Francisco: 5 m lidar terrain, real land cover, the real paths, buildings and Rec Center

Real weather

Daily rain from Oct 2020 to Sept 2023: a 230 mm drought year, then the New Year's Eve 2022 storm, 139 mm in one day

Your choices

Plant, water, keep or reroute a trail, build a rain garden. A teaching model computes what survives and what reaches the drain.

Same weather for everyone; different choices, different endings. That is what makes the lesson stick.

REAL DATA

Every layer of the island is public data

Layer	Source	What it drives in the game
Terrain, 5 m lidar bare earth (2023)	USGS 3DEP	The island’s shape and where water runs
Land cover, 10 m (2021)	ESA WorldCover	Golden grassland, trees, turf, paving
Paths, trails, lawns, named places	OpenStreetMap	The shortcut, the Rec Center, field notes
Buildings and park boundary	DataSF	The neighbourhood around the park
Soils	USDA NRCS SSURGO	How fast rain soaks in
Daily rain, Oct 2020 to Sept 2023	NOAA GHCN-Daily, downtown SF	Every replayed day, the 2021 drought
Hourly rain, 31 Dec 2022	NOAA ISD, OAK and SFO airports	The storm, hour by hour
Reference evapotranspiration	Open-Meteo ERA5 (model output)	How fast the soil dries out
2006 management zones	SF Recreation and Park (historical)	Where grassland restoration was planned
Aerial photo 2024, satellite greenness	USDA NAIP, Sentinel-2	Optional map layers, display only

Rain station: USW00023272, 3.7 km from the park · all sources listed in the game’s About screen

THE MODEL

A small teaching model computes every outcome

Seedlings

A daily soil-water balance (FAO-56 style) on the real rain. Survival comes from the days of severe thirst in the first year.

Storm runoff

Each 5 m cell makes runoff hour by hour (SCS curve number). Water follows the steepest path down the lidar to a garden or the drain.

Gardens and tanks

Buckets that fill, overflow, soak in and drain. Water let out later through a pipe still counts as reaching the drain.

Land check

Inspired by Land Degradation Neutrality: each kind of land against doing nothing, no offsetting between kinds. Erosion is a relative index.

Not modelled: flooding downstream, pipes, groundwater, fire, species or animal numbers. Tests check that every ending can be reached.

GAME MODE

Get ready, survive the year, face the storm

1 · Autumn
Walk, plant, talk to neighbours, build

2 · The year
Real weather plays; water when they're thirsty

3 · Result
What survived, a level-up, one lesson

The storm
31 Dec 2022, hour by hour; call the peak

Steps 1–3 repeat for three autumns: 2020, 2021, 2022. Then 8 endings and 3 stars.

Frostpunk: a countdown to a great storm

Plants vs. Zombies: teach inside level 1

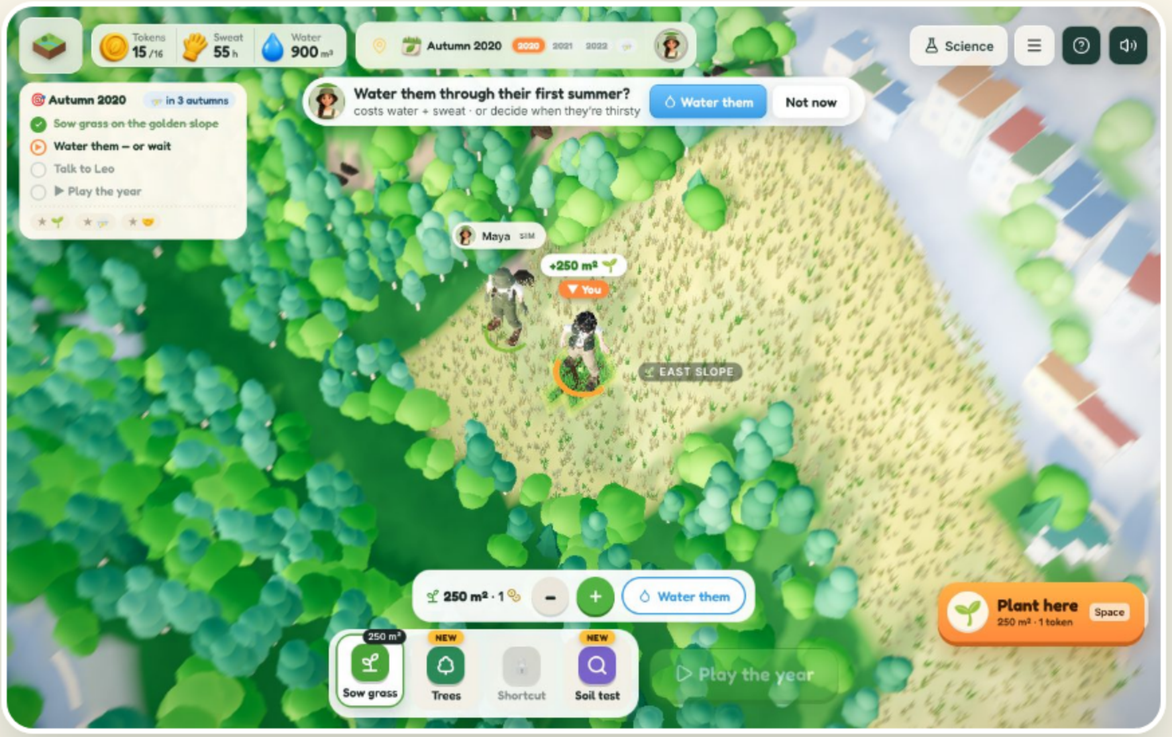
Animal Crossing: a guide and chained tasks

Terra Nil: progress is the land turning green

HOW IT HOOKS YOUNG PLAYERS

Built for a 15-year-old's first minute

- **The boss first:** a storm flash-forward, then a rewind to 2020
- **You are someone:** walk your character, plant with your own hands
- **Never lost:** a task list, one hint line, a go-here arrow
- **One big button** that changes with where you stand
- **Instant feedback:** sprouts, pops, level-ups, confetti
- **Collect and replay:** 18 badges, island decorations, field notes, 8 endings
- **Phone-ready,** no install, 6–8 minutes

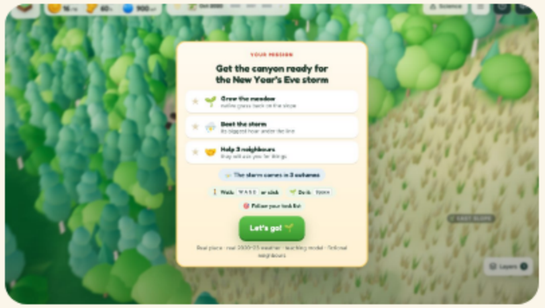


THE PLAYER JOURNEY

From the first tap to the ending in 6–8 minutes



1 · The storm first, then a rewind to 2020



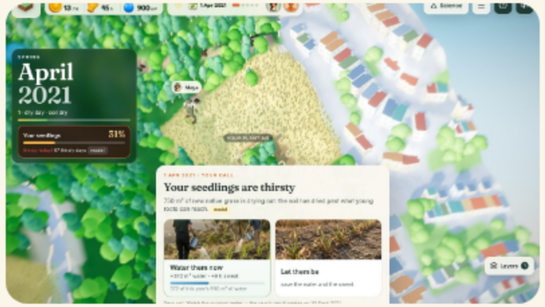
2 · Your mission: three stars



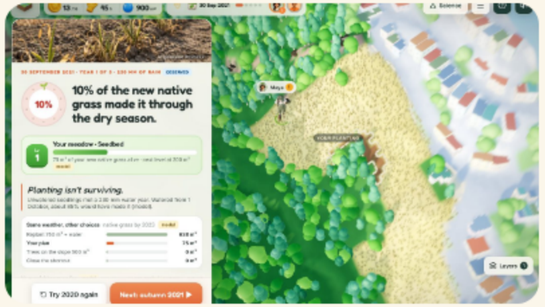
3 · Walk and plant with your own hands



4 · Leo waits on the shortcut: go and talk



5 · The drought year: water them or not?



6 · Result, level-up, one lesson



7 · New Year's Eve: call the peak hour



8 · The ending and your stars

PEOPLE, NOT JUST PLANTS

Four neighbours, four asks that pull against each other



Maya, volunteer: bring the native grass back on the slope

Leo, day-camp counselor: shade for his campers, keep their way down

Coach Rivera: save roof water for the field

Ms. Chen, downhill: less water rushing to the drain in the storm

The puzzle: closing the shortcut helps Maya but cuts off Leo. Rerouting it serves both. Help 3 of the 4 to earn a star.

WHAT PLAYERS LEARN

Every lesson comes from a choice you watch play out

You do	You see, in real weather	The lesson
Sow native grass	Covered ground holds; bare soil washes on storm days	Cover is the first defence
Water, or wait	In the 2021 drought most unwatered seedlings die	Timing is water
Plant trees on the grassland	Greener slope, smaller grassland	Restore the right land
Keep, close or reroute the trail	A trampled path stays bare; a reroute keeps a way down	People and land both count
Build a rain garden or tank	The storm’s peak hour drops; held water still drains later	Slow the peak; later isn’t less
Guess before a year	The model checks your guess	Predict, then test with data

The land check borrows the idea of Land Degradation Neutrality: avoid, reduce, reverse, and never offset one kind of land with another.

EIGHT ENDINGS

The model picks your ending from the slope, the storm and your neighbours

Neutral Ground
Grass back, storm under the line, 3+ neighbours helped

Trade-offs
Grass and storm done, fewer than 3 neighbours

Grassland Returns
The grass came back; the storm's peak got over the line

Storm Ready
Under the line at the Rec Center; the slope barely changed

The Peak Got Through
You built, but not where it caught the biggest hour

Greener, Not Better
Trees on the grassland: greener, and less grassland

The Long Dry
Sown in 2020, not watered, never sown again

Quiet Canyon
No moves at all; the canyon still changed

No random rolls. Every ending card is the same size: Neutral Ground is the challenge goal, not the only right answer.

HONEST BY DESIGN

Six rules we never break

No invented numbers

Every number comes from the model or observed data, tagged model or observed on screen

Only what we compute

No flooding, dust or water quality drawn, and no claims about animals or species

Fiction is labelled

The neighbours are simulated, and the ending pictures are marked as AI illustrations

Teaching tokens

Budgets are tokens, sweat hours and water, never real dollars

Inspired by LDN

The land check borrows the idea; it is never an official SDG 15.3.1 assessment

History is history

The 2006 management zones are a past plan, not a current approval or funding

HOW WE BUILT IT

A static web page that runs on any phone, even offline

Game

Vite, React 18 and TypeScript; three.js with React Three Fiber. One-file build of 13.5 MB that also plays offline.

Data and model

Public datasets on one 5 m grid of 211 × 280 cells. The model is plain TypeScript, with tests that reach every ending.

Art

Higgsfield: GPT Image 2.5 character art, Meshy 7 image-to-3D with auto-rigging, six shared moves. Slimmed with gltf-transform to about 2.8 MB.

Testing and AI help

Playwright playtests at laptop and phone sizes; independent review rounds; Claude for research, code, copy and bug audits.

RESEARCH AND ITERATION

What we read, and what players taught us

Research gathered

- UNCCD Land Degradation Neutrality: avoid, reduce, reverse, within the same kind of land
- SF Recreation and Park's 2006 natural areas plan for Glen Canyon's grassland
- NOAA records: the 2021 drought year and the New Year's Eve 2022 storm
- Youth climate survey (Lancet, 2021) and how hit games teach a first minute

What playtests changed

- Three independent review rounds scored 5, 6 and 6.5 out of 10
- "Looks like Google Maps, feels like a PowerPoint": a toy diorama, 3D characters and a story start
- "Where is this, and what do I do?": your own character, three stars, a task list and an arrow
- "One click skipped a year": a year starts only when your tasks are done

PLAY IT

Three autumns. One real storm. Your call.

Open the game link on a laptop or phone, or double-click the one-file version to play offline. A full season takes 6–8 minutes.

Next

- A classroom pilot with the How to Play guide
- More places: any park with lidar and daily rain records
- Translations, starting with Spanish and Chinese

